



QUARKUS CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & CLI

Installation

```
quarkus create app com.example:my-app
--extension=resteasy-reactive,hibernate-orm-p...
mvn quarkus:dev (live reload)
mvn package -Dnative (GraalVM native image)
quarkus extension add smallrye-health
```

04. PANACHE ENTITIES

Workflow

```
@Entity public class Item extends PanacheEnti...
public String name;
public static List<Item> findByName(String n){
    return list("name",n);
}
}
Item.findAll() / Item.count() / Item.listAll()
```

02. REST ENDPOINTS

Architecture

```
@Path("/items") @ApplicationScoped
public class ItemResource {
    @GET @Produces(APPLICATION_JSON)
    public List<Item> list(){ return Item.listAll...
    @POST
    @Transactional
    public void create(Item i){ i.persist(); }
}
```

05. REACTIVE PROGRAMMING

CRUD API

```
@GET @Path("/{id}")
public Uni<Item> getItem(@PathParam Long id){
    return Item.findById(id);
}
Multi<Item> stream = Item.streamAll();
<dep>quarkus-resteasy-reactive + hibernate-re...
```

03. CDI BEANS

YAML / Config

```
@ApplicationScoped / @RequestScoped / @Single...
@Inject MyService svc;
@ConfigProperty(name="greeting") String greet...
@Produces for custom beans
```

CDI context is compile-time no runtime scanning

application.properties for all config

06. NATIVE IMAGE BUILD

Integration

```
mvn package -Pnative (requires GraalVM)
mvn package -Pnative -Dquarkus.native.contain...
```

Results in `target/my-app-runner` binary

Startup <50ms; reduced memory footprint

Limitations: reflection, dynamic proxies need registration

@RegisterForReflection on needed classes



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. CONFIGURATION

Hardening

```
# application.properties
greeting=Hello
%prod.greeting=Hi there
@ConfigProperty(name="greeting",defaultValue=...
String greeting;
```

Promises: dev / test / prod auto-applied

10. MESSAGING (KAFKA)

Unit Tests

```
@Incoming("prices") void process(double price...
@Outgoing("prices-out") Multi<Double> generat...
<dep>quarkus-smallrye-reactive-messaging-kafk...
mp.messaging.incoming.prices.topic=prices-top...
mp.messaging.incoming.prices.bootstrap.server...
```

08. TESTING

Observability

```
@QuarkusTest
class ItemResourceTest {
@Test void testList(){
given().when().get("/items")
.then().statusCode(200);
}
}
@QuarkusIntegrationTest for native testing
```

11. COMMON GOTCHAS

Commands

CDI beans are resolved at build time runtime classloading fails
Native: reflection-heavy libs need @RegisterForReflection
Hibernate Reactive: can't mix blocking and reactive sessions
Dev mode hot reload works for config and code, not schema
Extension versions must match Quarkus BOM version

09. HEALTH CHECKS

Optimization

```
<dep>quarkus-smallrye-health</dep>
@Liveness / @Readiness
public class MyCheck implements HealthCheck{
public HealthCheckResponse call(){
return HealthCheckResponse.up("db");
}
}
GET /q/health/live /q/health/ready
```